

A1 Number — Answer Key

IGCSE Mathematics 0580 | Full Solutions

A1_Q1: Number Types — Problems

basic Unknown — Answer: $\sqrt{25} = 5$, which can be written as $\frac{5}{1}$. Therefore it is rational

basic Unknown — Answer: 1 , 7 , and $\sqrt{9} = 3$ are natural numbers. -2 is negative, 0 is not u

ermed Unknown — Answer: $\frac{22}{7}$ is rational because it is a fraction of two integers. It is appro

ermed Unknown — Answer: $\sqrt{2} \approx 1.414$ and $\sqrt{3} \approx 1.732$. A rational number between

dvance Unknown — Answer: Assume $\sqrt{2} = \frac{a}{b}$ where a, b are integers with no common factors

dvance Unknown — Answer: $\sqrt{3 + \sqrt{4}} = \sqrt{3 + 2} = \sqrt{5}$. Since 5 is not a perfect squa

halleng Unknown — Answer: No. Counterexample: $\sqrt{2}$ and $-\sqrt{2}$ are both irrational, but their su

halleng Unknown — Answer: This is true. Proof by contradiction: Suppose $x + y$ is rational. Then $y = (x$

A1_Q2: Factors, Multiples & Primes — Problems

basic Unknown — Answer: The factors of 36 are: $1, 2, 3, 4, 6, 9, 12, 18, 36$.

basic Unknown — Answer: No, 51 is not prime. $51 = 3 \times 17$, so it has factors $1, 3, 17, 51$ — mo

termed Unknown — Answer: Prime factorisation: $24 = 2^3 \times 3$ $60 = 2^2 \times 3 \times 5$ Common

termed Unknown — Answer: Prime factorisation: $18 = 2 \times 3^2$ $24 = 2^3 \times 3$ Each prime to t

dvance Unknown — Answer: We need the LCM of 36 and 48 . $36 = 2^2 \times 3^2$ $48 = 2^4 \times 3$

dvance Unknown — Answer: Prime factorise 126 : $126 = 2 \times 63 = 2 \times 3^2 \times 7 = 2^1 \times$

hallenc Unknown — Answer: Assume the set of primes is finite: $\{p_1, p_2, \dots, p_n\}$. Consider $N = p_$

hallenc Unknown — Answer: $n^2 - 1 = (n-1)(n+1)$. For this to be prime, one factor must be 1 (since prim

A1_Q3: Fractions & Decimals — Problems

basic Unknown — Answer: Common denominator is 20 : $\frac{3}{5} + \frac{1}{4} = \frac{12}{20} + \frac{5}{20}$

basic Unknown — Answer: $\frac{2}{3} \times \frac{5}{7} = \frac{2 \times 5}{3 \times 7} = \frac{10}{21}$

termed Unknown — Answer: $\frac{3}{4} \div \frac{9}{10} = \frac{3}{4} \times \frac{10}{9} = \frac{30}{36}$

termed Unknown — Answer: Let $x = 0.6666\dots$ $10x = 6.6666\dots$ $10x - x = 6$ $9x = 6$ $x = \frac{6}{9}$

dvance Unknown — Answer: Let $x = 0.1232323\dots$ $10x = 1.232323\dots$ $1000x = 123.232323\dots$ $1000x$

dvance Unknown — Answer: $1\frac{2}{3} = \frac{5}{3}$. Multiplication first: $\frac{2}{5} \times \frac{3}{}$

challenge Unknown — Answer: We need $\frac{a}{b}$ such that $\frac{1}{2} < \frac{a}{b} < \frac{2}{3}$ and a

challenge Unknown — Answer: Note that $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. So the sum telescope

A1_Q4: Percentages — Problems

basic Unknown — Answer: 15% of $\$240 = \frac{15}{100} \times 240 = 0.15 \times 240 = 36$. Answer: $\$36$

basic Unknown — Answer: $\frac{18}{45} \times 100\% = \frac{2}{5} \times 100\% = 40\%$.

merged Unknown — Answer: Increase = $\$92 - 80 = 12$. Percentage increase = $\frac{12}{80} \times 100\% = 15\%$

merged Unknown — Answer: After a 20% discount, the price is 80% of the original. $0.80 \times \text{text}$

advance Unknown — Answer: $A = P(1 + \frac{r}{100})^t = 5000 \times (1.03)^4$ $(1.03)^4 = 1.12550881$ $A =$

advance Unknown — Answer: Increase = $\$29500 - 25000 = 4500$. Percentage increase = $\frac{4500}{25000} \times 100\%$

challenge Unknown — Answer: After 3 years at 15% annual depreciation: Value $= P \times (0.85)^3$. $P \times$

challenge Unknown — Answer: First discount: multiply by 0.80 . Second discount: multiply by 0.90 . Overall

A1_Q5: Rounding & Estimation — Problems

basic Unknown — Answer: (a) $\$47.4$ (the second decimal is $8 \geq 5$). (b) $\$47.39$ (the third decimal is

basic Unknown — Answer: (a) \$0.005\$ (first non-zero digit is \$4\$, next digit \$5 \geq 5\$, so round up). (

termed Unknown — Answer: Lower bound = \$15.55\$ cm, upper bound = \$15.65\$ cm. So \$15.55 \leq \text{true le

termed Unknown — Answer: \$48.7 \approx 50\$, \$3.12 \approx 3\$, \$0.196 \approx 0.2\$. \$\frac{50 \times 3}{0}

dvance Unknown — Answer: For \$8.4\$: LB = \$8.35\$, UB = \$8.45\$. For \$5.3\$: LB = \$5.25\$, UB = \$5.35\$. Lower

dvance Unknown — Answer: Distance: \$235 \leq d < 245\$ km. Time: \$2.5 \leq t < 3.5\$ hours. Greatest speed

hallen Unknown — Answer: Length: \$115 \leq l < 125\$ m. Width: \$75 \leq w < 85\$ m. Perimeter = \$2(l+w)\$. L

hallen Unknown — Answer: \$63.8 \approx 60\$ to \$1\$ s.f. (or we could use \$64\$). Actually, rounding \$63.8\$

A1_Q6: Ratios & Proportion — Problems

basic Unknown — Answer: HCF of \$24\$ and \$18\$ is \$6\$. \$24 \div 6 = 4\$, \$18 \div 6 = 3\$. Simplified ratio:

basic Unknown — Answer: Total parts: \$2 + 3 = 5\$. One part: \$45 \div 5 = 9\$. First share: \$2 \times 9 =

termed Unknown — Answer: \$y = kx\$, so \$20 = k \times 5\$, \$k = 4\$. \$y = 4x\$. When \$x = 12\$, \$y = 4 \times

termed Unknown — Answer: \$y = \frac{k}{x}\$, so \$8 = \frac{k}{3}\$, \$k = 24\$. \$y = \frac{24}{x}\$. When \$x =

dvance Unknown — Answer: Scale factor = \$\frac{14}{8} = 1.75\$. Flour: \$500 \times 1.75 = 875\$ g. Sugar: \$

advance Unknown — Answer: Actual distance = $8 \times 50000 = 400000$ cm. $400000 \div 100 = 4000$ m. \$400

halleng Unknown — Answer: Inverse proportion: more taps $\rightarrow$ less time. $6 \times 45 = k$, so $k = 270$. \$10

halleng Unknown — Answer: Total parts: $3 + 5 + 7 = 15$. One part: $15000 \div 15 = 1000$. Friend 1: $3 \times 1000 = 3000$. \$3000

A1_Q7: Indices & Standard Form — Problems

basic Unknown — Answer: $2^3 \times 2^4 = 2^{3+4} = 2^7 = 128$.

basic Unknown — Answer: $0.00035 = 3.5 \times 10^{-4}$ (move the decimal point 4 places right).

termed Unknown — Answer: $5^{-2} = \frac{1}{5^2} = \frac{1}{25}$.

termed Unknown — Answer: $27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = 3^2 = 9$.

advance Unknown — Answer: $\frac{6^4 \times 6^{-2}}{6^3} = \frac{6^{4+(-2)}}{6^3} = \frac{6^2}{6^3} = 6^{-1} = \frac{1}{6}$

advance Unknown — Answer: $(3 \times 4) \times (10^4 \times 10^{-2}) = 12 \times 10^{4+(-2)} = 12 \times 10^2 = 1200$

halleng Unknown — Answer: Write all as powers of 2: $8 = 2^3$, $4 = 2^2$, $16 = 2^4$. $\frac{(2^3)^4}{(2^2)^3} = \frac{2^{12}}{2^6} = 2^6 = 64$

halleng Unknown — Answer: $\frac{5.97 \times 10^{24}}{7.35 \times 10^{22}} = \frac{5.97}{7.35} \times 10^{24-22} = 0.812 \times 10^2 = 81.2$